Cutaneous Squamous Cell Carcinoma — Ji et al., 2020

Malignant cell-state manifold · embryonic-program reproduction · CCCA meta-atlas

GO / NO-GO GATES

Gate 1 — Integration & continuity — GO

Patients thoroughly mixed after Harmony (theta=5, chosen on the integration sweep — mixing visually clean, 10x precedent); one connected component (99.7% of cells). Judged on the embedding, not a scalar.

Gate 2 — Embryonic module geography — GO (non-branching)

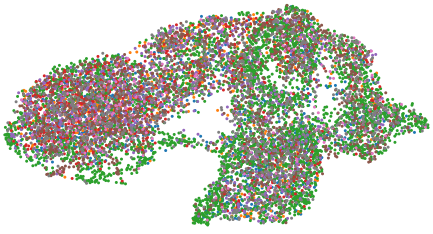
Gast & pSom overlap in the differentiating body (spatial r=+0.10) and both oppose Crest (r=-0.68 / -0.65); along pseudotime Gast declines (rho=-0.62) and Crest rises to a terminal neural-crest / TSK edge (Crest-high decile pt=0.79). One shared developmental axis, not two competing arms.

Gate 3 — Hub cross-repression — NA (branching-only)

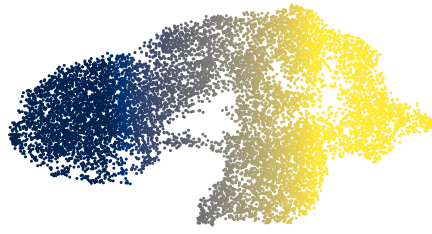
No bifurcation: the only PPT fork is the cycling compartment; the developmental axis is a single fold-free curve. Gate 3 not applicable. Organized gene-level Gast<->pSom cross-repression (r=-0.53) IS present on the hub, but with Gast/pSom overlapping and no arm resolution this is the non-branching-with-cross-repression case (task §1) — not a bifurcation.

STEP FIGURES

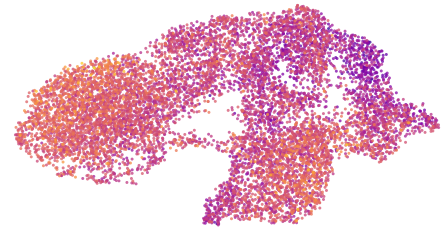
Samples (patient mixing)



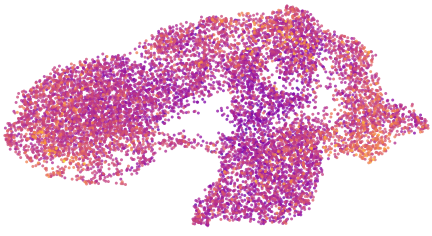
Pseudotime manifold



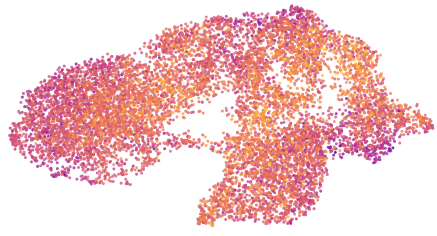
Gastrulation signature



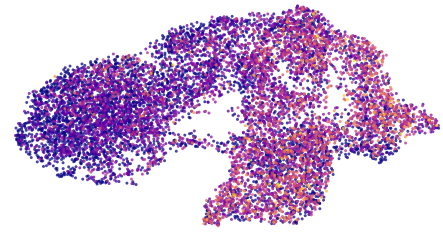
Somitogenesis signature



Neural-crest signature



EMT-TF: SNAI2 (terminus)



Dataset. Cutaneous squamous cell carcinoma (10x); 10,504 malignant cells from 8 patients (P3/P5 dropped below the count cliff).

Verdict. The embryonic pattern reproduces as a NON-BRANCHING axis: a Gast/pSom-high differentiating origin grades to a neural-crest-high basal/TSK terminus (Crest the inverse/terminal readout). No developmental bifurcation — the sole fork is proliferation. Linear geometry; Gate 3 not applicable.

PARAMETER CHOICES

Malignant load	10,598 cells / 10 patients (publication calls)
Sample threshold	>=103 (geometric mean of 268->40 count cliff); 8/10 kept -> 10,531 cells
Island removal	UMAP radius-graph (6x median NN dist); ~27 detached cells -> 10,504
Integration	Harmony on sample, theta=5 (chosen on integration sweep; 10x precedent); n_pcs=20; UMAP spread=1.3
Trajectory	scFates curve (ElPiGraph), Nodes=10, epg_mu=100; 2 tips, 0 forks; fold-free (self-approach=3.0)
Root	Gast/pSom-high, Crest-low curve tip (developmental origin)
Geometry	Linear (non-branching) — single continuous developmental axis
Hub (context)	Mixed+Diff differentiating core; gene-level Gast<->pSom r = -0.53
EMT-TFs	panel scored along pseudotime; monotone axis -> SNAI2 (terminus); PRRX2 root-end near floor
Annotation	publication cell_subtype_level1 + data-derived Mixed hub / Cycling
Status	Gate1 GO / Gate2 GO / Gate3 NA — pattern reproduces (non-branching)